ARCHITECTURAL DESIGN

OVERVIEW

Applying leadership and 21st century skills, participants develop a set of architectural plans and related materials in response to an annual architectural design challenge and construct a study model, as well as a computer-generated model and virtual walk-through, to accurately depict their design. Participants must demonstrate an understanding of and aptitude for architectural design, the development of plans, modeling techniques and practice, and the awareness of the role that the built environment can play in human behavior and interactions. The design problem for the current school year will be posted on the TSA website under *Themes & Problems*.

ELIGIBILITY

Two (2) teams of two to four (2-4) members per state may participate; individual entries are permitted.

TIME LIMITS

PRELIMINARY ROUND

A. All components of the chapter's entry (walk-through demonstration video URL and documentation portfolio PDF) must be finished, submitted, and accessible via the Internet by 11:59pm ET on a designated date mid-May.

SEMIFINAL ROUND

A. Up to twenty (20) minutes for the presentation/interview broken down as follows:
   1. Three (3) minutes for setup
   2. Ten (10) minutes for the presentation
   3. Five (5) minutes to respond to questions from the judges
   4. Two (2) minutes for breakdown

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.
B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.
C. Participants access the annual design challenge found on the TSA website under *Themes & Problems*.
D. Participants prepare a virtual walk-through demonstration video.
E. Participants prepare a documentation portfolio.
F. Participants prepare their study model.
G. Walk-through demonstration video URL and documentation portfolio PDF must be submitted by 11:59pm ET on a designated date in mid-May.
H. Submission information will be provided on the TSA website under *Competition Updates*.

PRELIMINARY ROUND

A. No more than (2) two team members report to the event area at the time and place stated in the conference program to submit the study model entry. All teams must turn in a study model on site.
B. Study Models are evaluated by judges. Neither students nor advisors are present at this time. The models judged will be the top twenty-four (24) selected from the results of the virtual walk-through demonstration video and the documentation portfolio.
C. A list of twelve (12) semifinalists in random order is posted.

SEMIFINAL ROUND

A. Semifinalists report at the time and place stated in the conference program to sign up for a presentation/interview.
B. Up to four (4) representatives from each semifinalist team report at the assigned time and place to participate in the presentation/interview.

C. One to four (1-4) members present the study model and participate in judges interview.

D. The top ten (10) finalists are announced at the awards ceremony.

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRE-CONFERENCE

A. Documentation Portfolio:

1. Documentation materials (comprising a "portfolio") are required and must be submitted as a multi-page PDF document with pages in this order:
    a. Title page with the event title, the team identification number, the conference city and state, and the year; one (1) page
    b. Table of contents; pages as needed
    c. A description of the individual/team's interpretation of the design challenge and an explanation of the style and merits of the design concepts; maximum of two (2) pages
    d. List and description of each of the construction systems (any and all that apply) and their incorporation and application to the solution: delivery, construction methods and materials, electrical wiring, plumbing, HVAC, and site requirements; maximum of six (6) pages
    e. A schedule of finish materials for all exterior and interior surfaces of the architectural design (this is not a list of the model construction materials); one (1) page
    f. A complete set of student created drawings (can be CAD drawings) appropriate scaled to fit the PDF format required for submission.
        i. Drawings; pages as needed
            1. Site Plan
            2. Overall Floor Plan(s)
            3. Enlarged Floor Plans as required to describe design elements
            4. Roof Plan
            5. Exterior Elevations
            6. Building Section(s)
            7. Interior Elevation(s) or Perspective(s)
        ii. A 3D modeling/rendering drawing of the individual/team's final design with appropriate details included; one (1) page
    g. Photographs of the study model (maximum of four photos per page); maximum of three (3) pages
    h. List of resources/references; pages as needed
    i. Photo/Film/Video Consent and Release Forms – recognizable individuals pictured in the images (minors require parental consent) must give their written consent before the images can be used in this event (see 2027 & 2028 Forms Appendix); pages as needed
    j. Resources and AI Reflection (see 2027 & 2028 Forms Appendix); maximum of four (4) pages
2. A PDF copy of the documentation portfolio is submitted pre-conference. A printed copy of the documentation portfolio, secured in a clear front report cover with single sided pages, is submitted on-site with the study model in the preliminary round.

B. Virtual Walk-through Presentation Video; maximum two (2) minutes.

1. Video of virtual model presented by student with description highlighting the design attributes.
2. Participants may choose a video hosting site (such as an UNLISTED YouTube URL), or a shareable link in cloud storage, as long as the video is located on-line and accessible for evaluation. Any video link that requires permissions or a download will not be judged.

C. Model:
   1. Study model must be placed on a site board no larger than 20" x 20".
   2. Study model construction concepts, materials, techniques, and applications:
      a. Foam core sheet or similar materials are suggested (but not limited to) for use as interior walls, exterior walls, and roof construction.
      b. Foam core board that is ½" thick or greater is recommended for use as the site board for the model.
   3. Teams should submit a printed copy of their portfolio for judges to use as reference during the judging of the study model.
      a. printer portfolio should be secured in a clear view report cover with single sided pages.
      b. the portfolio must be incorporated into the study model display not to exceed the size of 20" x 20".

PRELIMINARY ROUND

A. No more than two (2) team members report to the event area at the time and place stated in the conference program to submit the study model.
B. Study Models are evaluated by judges. Neither students nor advisors are present at this time. Study models judged will be the top twenty-four (24) selected from the results of the virtual walk-through demonstration video and the documentation portfolio.
C. A list of twelve (12) semifinalists (in random order) is posted.

SEMIFINAL ROUND

A. Semifinalists report at the time and place stated in the conference program to sign up for a presentation/interview.
B. Up to four (4) representatives from each semifinalist team report at the assigned time and place to participate in the presentation/interview.
C. One to four (1-4) members present the study model and participate in judges interview.
D. A table will be provided for the study model presentation.

EVALUATION

PRELIMINARY ROUND

Tier 1

A. Virtual walk-through demonstration video
B. Documentation portfolio

Tier 2

C. Study model

SEMIFINAL ROUND

A. Study model presentation/interview

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Appraiser
- Architect
- Construction manager
- Interior designer
- Urban and regional planner

Participant/Team ID# _____

ARCHITECTURAL DESIGN

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ TIER 1 – Virtual walk-through demonstration video is uploaded and playable
☐ TIER 1 – Documentation portfolio is present
☐ TIER 2 – Study model is present
☐ ENTRY NOT EVALUATED

TIER 1 – VIRTUAL WALK-THROUGH DEMONSTRATION VIDEO (60 points)				
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	Record scores in the column spaces below.
	1-4 points	5-8 points	9-10 points	
Virtual Model Design (X2)	Model lacks detail, accuracy, or completeness; key architectural elements are missing or poorly represented; scale, proportions, or spatial relationships are inaccurate; design does not clearly reflect the challenge requirements.	Model includes most major architectural elements; scale and proportions are generally accurate; design reflects the challenge but lacks refinement or depth; some details or spatial relationships may be unclear.	Model is highly detailed, accurate, and complete; scale and proportions are precise and realistic; design fully reflects and enhances the challenge requirements; spatial relationships and architectural elements are clearly and effectively represented.	
Design Attributes Explained (X2)	Little or no explanation of design choices; connections to the design challenge are unclear; architectural terminology is missing or misused.	Basic explanation of design choices is provided; some connection to the design challenge is evident; uses some appropriate architectural terminology.	Clear, thorough explanation of design decisions and features; strong connection between design and challenge requirements; effective use of architectural terminology and concepts; demonstrates understanding of how design impacts function and users.	
Voice Over Content and Quality (X1)	Voice is unclear, too quiet, or difficult to understand; limited or disorganized explanation; frequent pauses, filler words, or reading directly from script.	Voice is generally clear and understandable; explanation is somewhat organized; minor issues with pacing, clarity, or delivery.	Voice is clear, confident, and easy to understand; explanation is well-organized and engaging; smooth pacing with minimal filler or hesitation; enhances viewer understanding of the design.	
Video Length (X1)	Video significantly exceeds or falls short of the time limit; important content is missing or rushed due to poor timing.	Video is close to the required time limit; minor pacing issues; some sections may feel rushed or extended.	Video stays within the required time limit; time is used effectively to present all key aspects of the design; smooth pacing with balanced coverage of content.	
			TIER 1 – VIRTUAL WALK-THROUGH DEMONSTRATION VIDEO SUBTOTAL (60 points)	

TIER 1 – DOCUMENTATION PORTFOLIO (110 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Components (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Description of Design Interpretation (X1)	Description of the design and style is unclear or vague.	Description of the design and explanation of the style are included; they are adequately presented.	Description and merits of the design and explanation of the style are clear, effective, and convincing.	
Construction Systems (X1)	There is little or no evidence of attention to the various construction systems.	Most, but not all, construction systems are addressed; they are generally well presented.	All applicable construction systems are addressed, clearly documented, and well presented.	
Schedule of Finish Materials (X1)	Many elements of the interior and exterior finish schedules are missing or incomplete.	Most, but not all, elements of the interior and exterior finish schedules are included.	All interior and exterior finish schedules/materials are detailed and explained clearly.	
Drawings (X2)	A few of the required drawings are present, but they are lacking in quality.	Most, but not all, of the required drawings are included and are in the proper format.	All required drawings are included and in the proper format.	
3D Modeling/ Rendering (X1)	Drawing is present and displays minimal effort.	Drawing is complete, clear, and includes details.	Drawing is high quality, clear, and includes necessary details.	
Model Presentation Documents (X1)	Hyperlink to the walk-through video may be included but not working or not included at all; photo/video release and consent forms are not included.	Hyperlink to the walk-through video is included and active, but the photo/video release and consent forms are not included.	Hyperlink to the walk-through video is included and active; all photo/ video release and consent forms are included.	
Resources/ References (X1)	There is little or no effort to provide resources and references.	Resources and references included are generally presented appropriately.	There is clear evidence of the appropriate use of applicable resources and references.	
Photographs of the Study Model (X1)	Photographs only show simple views of the model and are of poor quality.	Photographs show multiple views from various angles and the quality is good.	Photographs are high quality and show multiple views and angles of the model's interior (with roof removed, if applicable) and exterior.	
Resources and AI Reflection (X1)	Form is incomplete or vague and does not clearly explain whether or how AI was used, showing little connection to the submission or understanding of effectiveness and ethics.	Form is complete and clearly explains AI use or the lack thereof with connections to the submission or theme/topic (when applicable), addressing effectiveness and ethical considerations with adequate analysis.	Form is complete, concise, and specific, critically evaluating how AI or non-AI tools shaped decisions and outcomes, with strong alignment to the submission or theme/topic (when applicable) and ethical reasoning.	
			TIER 1 – DOCUMENTATION PORTFOLIO SUBTOTAL (110 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

TIER 1 – SUBTOTAL (170 points)	

TIER 2 – STUDY MODEL (100 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Effectiveness of Design (X2)	Design is ineffective in meeting the needs of the challenge.	Design is somewhat effective in meeting the needs of the challenge.	Design is clearly effective in meeting the needs of the challenge.	
Access and Flow (X1)	Design reflects an ineffective traffic flow pattern and/or use of space to gain access to the structure.	Design reflects a somewhat effective traffic flow pattern and use of space to access the structure.	Design presents a clear, effective traffic flow pattern and full consideration of the use of space.	
Creativity and innovation (X2)	Design lacks originality and exhibits few, if any, creative and/or innovative applications.	Some unique, innovative, and creative concepts are incorporated in the overall design.	Unique, creative, and innovative approaches are fully incorporated into the design.	
Quality of Construction (X2)	Construction is of poor quality and appearance, with little or no attention to neatness.	Construction is somewhat neat and has appropriate quality and appearance.	Construction is of excellent quality and exemplary appearance.	
Use of Materials (X1)	Choice of materials is ineffective and inadequate for the type and scale needed.	There is effective choice of materials and some attention to scale.	There is effective and excellent use of materials and accurate choice of scale.	
Design Representation (X2)	Model is ineffective in depicting the requirements of the design challenge.	Model is somewhat effective in depicting the requirements of the design challenge.	Model clearly and effectively incorporates and depicts all aspects of the design challenge.	
			TIER 2 – STUDY MODEL SUBTOTAL (100 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (270 points)

SEMIFINAL PRESENTATION/INTERVIEW (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Organization of Presentation (X1)	Participant(s) seems unorganized and unprepared for the presentation.	Participant(s) is/are generally prepared/organized in the overall presentation.	Presentation is logical, well organized, and easy to follow.	
Knowledge (X2)	Participant(s) seems to have little understanding of the concepts of the design challenge; vague answers to interview questions are provided.	An understanding of the concepts of the design challenge, and answers to questions, are adequate.	There is clear evidence of a thorough understanding of the design challenge; questions are answered well.	
Articulation (X1)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21st century skills are clearly evident.	
Delivery (X1)	Team/individual is verbose and/or uncertain in the presentation/ interview; participant posture, gestures, and lack of eye contact diminish the delivery.	Team/individual is somewhat well-spoken and clear in the presentation/interview; participant posture, gestures, and eye contact result in an acceptable delivery.	Team/individual is well-spoken and distinct in the presentation/interview; participant posture, gestures, and eye contact result in a polished, natural, and effective delivery.	
			SEMIFINAL PRESENTATION/INTERVIEW SUBTOTAL (50 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

SEMIFINAL SUBTOTAL (50 points)

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL** (320 points)

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

ARCHITECTURAL DESIGN
EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator

B. Assistants for check-in, two (2)

C. Judges:

1. Two (2) or more (virtual walk-through demonstration videos and documentation portfolios)
2. Two (2) or more (study models)
3. Two (2) or more (study model and semifinal presentation/interview; preferably the same judges who reviewed the documentation portfolios)

MATERIALS

A. Coordinator's packet, containing

1. Event guidelines, one (1) copy for the coordinator and for each judge
2. TSA Event Coordinator Report
3. Stick-on labels for entries, as needed

B. Measuring tape

C. Tables for study model check in

D. Table for the study model during the semifinal round

E. Tables and chairs for judges

RESPONSIBILITIES

PRE-CONFERENCE

A. National TSA will collect entries until 11:59pm ET on a designated date in mid-May. The results will be shared with the CRC manager, event coordinator, and assigned judges.

B. Review entries as they are submitted to the designated online storage utility.

C. Manage communication and Pre-conference evaluation (at least two [2] or more judges should be recruited earlier in the year). Coordinate with National TSA and/or the Judge Manager.

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designate time and location.

B. Report to the CRC room and check the contents of the coordinator's packet.

C. Review the event guidelines and check to see that enough personnel have been scheduled.

D. Inspect the area or room in which the event is being held for appropriate set-up, including room size, chairs, tables, outlets, etc. Notify the event manager of any potential problems.

E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.

F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

A. Check-in the study models at the time and place stated in the conference program.

B. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.

C. Each entry must include the team's identification number in the upper right-hand corner of the entry.

D. Secure the entries in the designated area.

PRELIMINARY ROUND

A. Judges independently evaluate the top twenty-four (24) study models to determine the twelve (12) semifinalists.

B. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:

   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

C. Create a sign-up sheet for semifinalist presentation/interviews.

SEMIFINAL ROUND

A. Semifinalists report at the time and place stated in the conference program to sign up for a presentation/interview time.

B. Semifinalists report at the assigned time and place for the presentation/interview.

C. Manage completion of the on-site presentation/interviews.

D. Discuss rule violations (e.g. 20% deduction, disqualification) and have all relevant parties initial the rating form.

E. If necessary, manage security and the removal of materials from the area.